Practical Name: AI-Powered Software Testing

Aim: To analyze AI-powered software testing tools and apply Generative AI techniques for test case generation, test data creation, defect prediction, requirement analysis and automated Selenium test script generation for software quality assurance.

Source Code

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

driver = webdriver.Chrome()
driver.maximize_window()

try:
    driver.get("https://www.saucedemo.com/")
    wait = WebDriverWait(driver, 10)

    username = wait.until(
        EC.visibility_of_element_located((By.ID, "user-name"))
    )
    password = driver.find_element(By.ID, "password")
    login_button = driver.find_element(By.ID, "login-button")

    username.send_keys("standard_user")
    password.send_keys("secret_sauce")
    login_button.click()

    wait.until(EC.url_contains("inventory"))
    assert "inventory" in driver.current_url

    print("AI-Assisted Selenium Test: PASS")
    print("Requirement Validation: PASS")

except AssertionError:
    print("AI-Assisted Selenium Test: FAIL")

finally:
    driver.quit()
```